

CSE 342
Statistical Machine Learning
Assignment - 2

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Code Explanation :

The code is divided into functional components, with each function performing a specific task in the overall pipeline. Experimentation is performed by setting a Random Seed (**2023115**) to ensure reproducible results across iterations.

- ***load_dataset():*** This function loads the MNIST dataset by taking file paths of labels and images stored in ubyte format and performs processing using struct, and returns numpy arrays of images and labels.
- ***preprocess():*** This function takes input of numpy arrays of images and flattens the 2D image array into a 1D array and normalises the values into the range [0,1], and returns the normalised image array
- ***sample_data():*** This function is used to sample data points with count specified as sample_size and after sampling adds to individual image array and label array and shuffles them using numpy permutation and returns the shuffled numpy arrays of the data points.
- ***filter012():*** This function filters out training images and labels that belong to classes 0, 1, and 2. And returns the updated subset of images and labels.
- ***MLEstimate():*** This function estimates the parameters for a Gaussian-based classifier, like Mean, Covariance, and prior, from the input training images and labels, and returns a list of ordered parameters (mean, cov, prior) for each class c.
- ***LDA():*** This implements Linear Discriminant Analysis for classification. It assumes all classes share a single "pooled" covariance matrix and calculates a discriminant score for each class based on the distance between the test image and the class mean; the class with the highest score is chosen as the prediction.
- ***QDA():*** This implements Quadratic Discriminant Analysis. Unlike LDA, it uses a unique covariance matrix for each class, allowing more flexible

decision boundaries; it computes the log-determinant and inverse of each class's covariance to determine the most likely class for a given image.

- ***principalComponent()***: This function is a Principal Component Analysis (PCA) of a centered matrix of data, train images (e.g., features are image pixels). It calculates the mean-subtracted covariance matrix, derives and ranks the real eigenvalues/eigenvectors by decreasing variance, and selects k items using either the comp parameter or the cumulative-variance criterion var. Lastly, it projects the data onto the top k eigenvectors, and the reduced data Y, the projection matrix Up, and the top eigenvalues are returned.
- ***fischerDA()***: This function is a dimensionality reduction using Linear Discriminant Analysis (LDA) as suggested by Fisher on 3-class train images. It estimates the global mean, within-class scatter matrix Sw, and between-class scatter Sbw (weighted class mean differences), and regularizes Sw to invert it. The eigenvectors of the generalized eigenvalue problem $\text{inv}(\mathbf{S_w}) @ \mathbf{S_{bw}}$ are used to obtain the top 2 real-valued directions of W, which maximize the separability of the classes.

FDA with LDA

Train Accuracy:

LDA Discriminant values for first test sample: ['Class 0: -1.10', 'Class 1: -14767.49', 'Class 2: -10301.65']

LDA Accuracy: 1.0000

Test Accuracy:

LDA Discriminant values for first test sample: ['Class 0: -9492.84', 'Class 1: -2118.92', 'Class 2: -1551.79']

LDA Accuracy: 0.8667

FDA with QDA

Train Accuracy:

QDA Discriminant values for first test sample: ['Class 0: 10.41', 'Class 1: -14755.97', 'Class 2: -10290.14']

QDA Accuracy: 1.0000

Test Accuracy:

QDA Discriminant values for first test sample: ['Class 0: -9481.32', 'Class 1: -2107.41', 'Class 2: -1540.28']

QDA Accuracy: 0.8667

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PCA 75% Variance + LDA
(20,)
(20,)
(20,)
Train Accuracy:
LDA Discriminant values for first test sample: ['Class 0: -18.10', 'Class 1: -40.02', 'Class 2: -26.25']
LDA Accuracy: 0.9667
Test Accuracy:
LDA Discriminant values for first test sample: ['Class 0: -23.70', 'Class 1: -17.12', 'Class 2: -8.24']
LDA Accuracy: 0.9400

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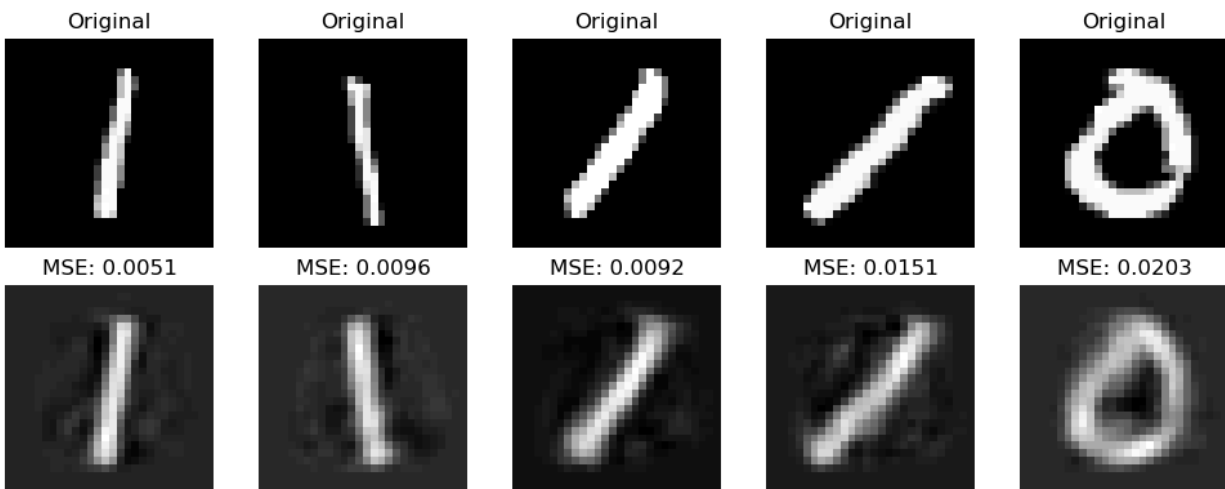
PCA 90% Variance + LDA
Size of Data Matrix (784, 300)
(51,)
(51,)
(51,)
Train Accuracy:
LDA Discriminant values for first test sample: ['Class 0: -49.35', 'Class 1: -88.11', 'Class 2: -65.68']
LDA Accuracy: 0.9767
Test Accuracy:
LDA Discriminant values for first test sample: ['Class 0: -50.33', 'Class 1: -43.15', 'Class 2: -31.10']
LDA Accuracy: 0.9333

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PCA First 2 Components + LDA
Size of Data Matrix (784, 300)
(2,)
(2,)
(2,)
Train Accuracy:
LDA Discriminant values for first test sample: ['Class 0: -2.56', 'Class 1: -8.32', 'Class 2: -4.37']
LDA Accuracy: 0.9033
Test Accuracy:
LDA Discriminant values for first test sample: ['Class 0: -13.43', 'Class 1: -4.38', 'Class 2: -1.64']
LDA Accuracy: 0.9167

```

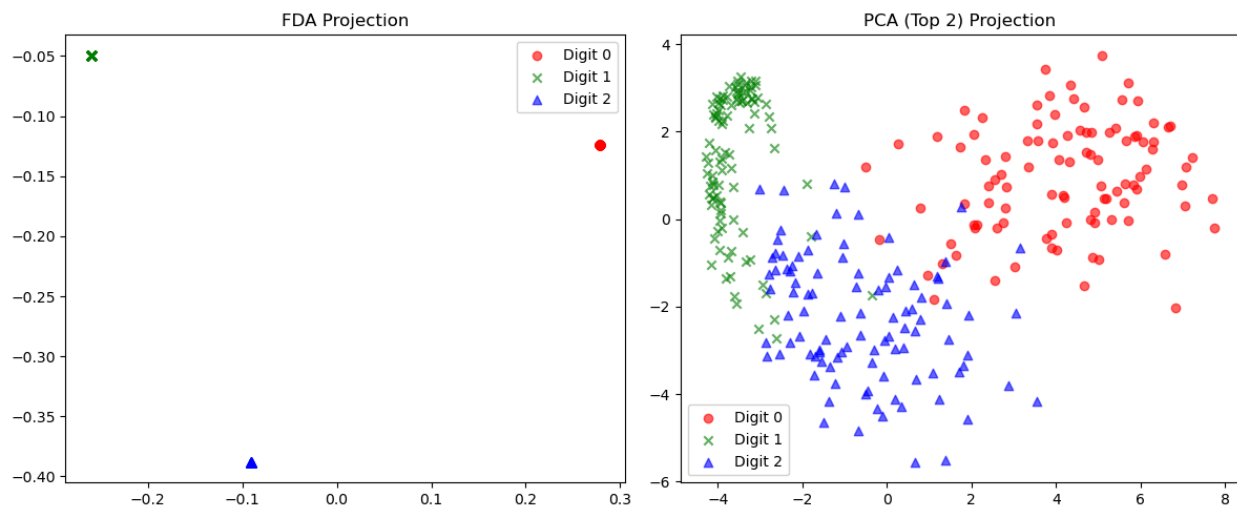


Analysis of how PCA affects classification performance.

Maximal Variance: The optimal variance to retain is 75% (i.e., 20 components), yielding the maximum test accuracy (94.00%). It was a good dimensional reduction method that maintained the main structural details of the digits.

Over-retention: When variance was increased to 90, there was a slight improvement in training accuracy (94.740) and a slight decrease in test accuracy (93.33). This indicates that the added components introduced a bit of noise or slight overfitting.

Under-retention (Top 2 Components): Training accuracy (90.33%) was reduced dramatically in case of reducing the data to two components only. Two dimensions are inadequate to generate the rich outlines of linear decision-making that would be very useful in this case.



Visualisation

In both the projected curves for the FDA and PCA Techniques, we observe that the FDA projection separates the clusters, as it maximizes the between-class scatter and minimizes the within-class scatter, thereby achieving optimal separability. Unlike in PCA, where the axes of maximum variance are determined to project the original data and thereby reduce the dimensions, we observe some overlap among classes rather than complete separability.